

IT-22: Software Testing and Quality Assurance				
Teaching Scheme: Theory Sessions: Total 45 Hours		Credit: 03	Examination Scheme: Internal (TH): 25 Marks External (TH): 50 Marks Total :75 Marks	
Prerequisites: Basic concepts software development				
Course Objectives: - To understand the principles of software development emphasizing processes and activities of quality assurance - To study fundamental concepts in software testing, including software testing objectives, process, strategies, and methods. - To understand test design techniques based on functionality and structure of software - To understand test planning, monitoring, and control process - To gain the techniques and skills on how to use software testing tools to support software testing activities				
Course Outcomes: On completion of the course, learners should be able to				
CO#	Cognitive Domain	Course Outcomes		
CO1	Understand	Understand the role of software quality assurance in contributing to the efficient delivery of software solutions.		
CO2	Understand	Understand specific software tests with well-defined objectives and targets.		
CO3	Apply	Apply the software testing techniques in commercial environments.		
CO4	Analyze	Construct test strategies and plans for software testing.		
CO5	Apply	Demonstrate the usage of software testing tools for test effectiveness, efficiency, and coverage.		
Unit No.	Content		Weightage in %	No of Sessions
1	Software Quality Assurance Fundamentals 1.1. Definition of Quality, Quality Assurance, Quality Control, Difference between QA and QC, SoftwareQuality Assurance Challenges 1.2. Software Quality Assurance, SQA Planning &Standards (ISO 9000) 1.3. SQA Activities 1.4. Building Blocks of SQA 1.5. Software Quality factors 1.6. Software Reliability & Reliability Measurement Factors: ROCOF, MTTF, MTTR, MTBF, POFOD, Availability		20	8
*Mapping of Course Outcomes for Unit 1: CO1				

2	Software Testing Fundamentals 2.1. Definition & Objectives of Testing 2.2. Role of testing and its effect on quality 2.3. Causes of software failure: Definition of -Error, Bug, Fault, Defect and Failure, 2.4. Economics of Testing 2.5. Seven Testing Principles 2.6. Software Testing Life cycle 2.7. Validation & Verification Concepts - V Model and WModel 2.8. Agile Testing- Test Driven Software Development 2.9. Levels of Testing- 2.9.1. Unit (Component) Testing 2.9.2. Integration Testing 2.9.3. System Testing 2.9.4. User Acceptance Testing (UAT) 2.10. Test Types 2.10.1. Functional testing (Black-box) 2.10.2. Non-functional testing (Testing of software product characteristics) 2.10.3. Structural testing (White-box) 2.10.4. Testing related to changes – Confirmation (Re-testing) and Regression Testing 2.11. Non-Functional Testing Types – 2.11.1. Performance (Load & Stress) 2.11.2. Usability 2.11.3. Maintainability 2.11.4. Portability 2.11.5. Security 2.11.6. Localization & Internationalization 2.12. Concept of Smoke testing and Sanity Testing	20	10
*Mapping of Course Outcomes for Unit 2: CO2			
3	Static & Dynamic Testing 3.1. Static Techniques – Review 3.1.1. Review Process (Informal & Formal) 3.1.2. Technical or Peer Review 3.1.3. Walkthrough 3.1.4. Inspection 3.2. Static Techniques – Static Analysis 3.2.1. Static Analysis by Tools (Automated Static Analysis) 3.3. Test Design Techniques: Black Box Testing 3.3.1. Equivalence Partitioning	20	9

	3.3.2. Boundary Value Analysis 3.3.3. Decision Table Testing 3.3.4. State Transition Testing 3.4. Test Design Techniques -White Box Testing Techniques (coverage based and fault-based) 3.4.1. Statement coverage 3.4.2. Branch & Decision coverage 3.4.3. Path coverage 3.4.4. McCabe's Cyclomatic Complexity Metric (Computation of Cyclomatic Complexity to be covered) 3.4.5. Data Flow based Testing 3.4.6. Mutation Testing 3.5. Test Design Techniques -Experience based techniques 3.5.1. Error Guessing 3.5.2. Exploratory Testing		
*Mapping of Course Outcomes for Unit 3: CO3			
4	Test Management 4.1. Test Organization- Roles & Skills of Tester, Test Lead, Test Manager 4.2. Test Planning- Test Plan as per IEEE 829 STANDARDTEST PLAN TEMPLATE 4.3. Test Process Monitoring & Control 4.3.1. Test Monitoring through -Test Log (IEEE 829: TEST LOG TEMPLATE) and Defect Density 4.3.2. Reporting Test Status (IEEE 829: TEST SUMMARY REPORT TEMPLATE) 4.3.3 Test Control 4.4. Test Scenario, Test Suite, Test Cases (bothPositive & Negative Test Cases, as per IEEE 829: TEST CASE SPECIFICATION TEMPLATE) 4.5. Configuration Management- ConfigurationManagement support for Testing 4.6. Risk and Testing- Project Risk & Product Risk 4.7. Incident/ Defect Management 4.7.1. Defect Life Cycle 4.7.2. Defect/ Incident Report (IEEE 829: TEST INCIDENT REPORT TEMPLATE) Case Study on Test Plan for applications and Case study onTest Cases for different features within applications	25	11
*Mapping of Course Outcomes for Unit 4: CO4			
5	Tool Support for Testing 5.1. Types of Test tools –CAST (only type &	15	7

	their purpose & Benefits and Risks should be covered) 5.2. Introduction of a tool into an organization 5.3. Testing tools 5.3.1. Selenium -WebDriver and Test NG 5.3.2. JMeter 5.3.3. Postman 5.3.4. ETL Testing Tool 5.4. JIRA (Project Management)		
*Mapping of Course Outcomes for Unit 5: CO5			
Learning Resources			
Text Books: - Foundations of Software Testing by Rex black, Erik Van Veenendaal, Dorothy Graham (2020)-Cengage Learning: London UK, 5th Edition - Software Engineering by Sommerville-Pearson, 8th Edition - Daniel Galin, "Software Quality Assurance: From Theory to Implementation", Pearson - Addison-Wesley, 2012. 2. - Effective Methods for Software Testing by William Perry- Wiley Pub, 3rd Edition.			
Reference Books: - Roger S. Pressman, "Software Engineering-A Practitioner's Approach", McGraw Hill - pub.2010 - Software Testing in Real World Edward Kit- Pearson Pub - Software Testing Techniques by Boris Beizer-DreamTech Pub, 2nd Edition - Software Testing by Ron Patton, TechMedia Pub. - Introducing Software by Testing Louise Tamres - Fundamentals of Software Engineering –Rajib Mall, 3rd Edition - Allen Gilles "Software quality: Theory and management", International Thomson, - Computer press 1997. - Software Testing Principles Techniques and Tools by Milind. G. Limaye- Tata McGraw Hill Pub. - Stephen H. Kan, "Metrics and models in software quality Engineering", Addison – Wesley 2003.			
Recommended Learning Material: www.istqb.org https://www.seleniumhq.org/ https://www.softwaretestingmaterial.com/selenium-tutorial/ https://www.toolsqa.com/selenium-tutorial/ www.guru99.com/software-testing.html www.guru99.com/selenium-tutorial.html www.guru99.com/mobile-testing.html www.professionalqa.com			

- www.resources.sei.cmu.edu/library
- www.iist.org

Recommended Certifications:

- CAST (Certified Associate in Software Testing)
- CSQA (Certified Software Quality Analyst Certification)
- (ISTQB) International Software Testing Qualifications Board Certification
- (CQE) Certified Quality Engineer
- (CMST) Certified Manager of Software Testing